

Radioactive Decay and Half-Life

Answer key with worked solutions

A. Understanding

1. Definitions.

(a) The spontaneous decay of **unstable nuclei**, which emit radiation (alpha, beta, positron or gamma). (b) The **time for half** of the radioactive nuclei in a sample (and its activity) to decay. (c) The decay of any one nucleus is **random**, but in a very large number of nuclei a constant **fraction** decays in each unit of time, so the sample as a whole follows the half-life.

B. Nuclei and nuclear equations

2. Protons and neutrons.

(a) **92 p, 143 n** (b) **27 p, 33 n** (c) **53 p, 78 n** (d) **86 p, 136 n**. Neutrons = $A - Z$.

3. Completed equations.

(a) ${}_{86}^{222}\text{Rn}$ (b) ${}_{54}^{131}\text{Xe}$ (c) ${}_{8}^{18}\text{O}$ (d) ${}_{28}^{60}\text{Ni}$ (e) ${}_{92}^{235}\text{U}$ (f) ${}_{20}^{40}\text{Ca}$. In each, the mass numbers and atomic numbers balance; for example (e): $239 = 235 + 4$ and $94 = 92 + 2$.

4. Identify X.

(a) $A = 0$, $Z = -1$: a **beta particle**, ${}_{-1}^0\text{e}$. (b) $A = 4$, $Z = 2$: an **alpha particle**, ${}_2^4\text{He}$. (c) $A = 0$, $Z = +1$: a **positron**, ${}_{+1}^0\text{e}$.

5. Uranium-238 series.

${}_{92}^{238}\text{U} \longrightarrow {}_{90}^{234}\text{Th} + {}_2^4\text{He}$; ${}_{90}^{234}\text{Th} \longrightarrow {}_{91}^{234}\text{Pa} + {}_{-1}^0\text{e}$; ${}_{91}^{234}\text{Pa} \longrightarrow {}_{92}^{234}\text{U} + {}_{-1}^0\text{e}$. Final nucleus: **uranium-234**, ${}_{92}^{234}\text{U}$ (an isotope of the starting element).

C. Types of radiation

6. Comparing radiation.

Alpha: a helium nucleus, charge +2; stopped by paper or skin. **Beta:** an electron, charge -1; stopped by a few mm of aluminium. **Gamma:** electromagnetic radiation, no charge; only reduced by thick lead or concrete. Inside the body, an alpha emitter deposits all its energy in a very short distance in living cells, and alpha particles ionize strongly, so they cause the most damage there.

D. Half-life

7. Technetium-99m.

$$n = \frac{24.04 \text{ h}}{6.01 \text{ h}} = 4.00 \text{ half-lives}$$
$$\text{activity} = 800 \text{ MBq} \times \left(\frac{1}{2}\right)^{4.00} = \frac{800 \text{ MBq}}{16} = \mathbf{50.0 \text{ MBq}}$$

8. Half-life from data, and radon.

(a) $1600 \text{ Bq} \rightarrow 800 \text{ Bq} \rightarrow 400 \text{ Bq}$ is **2 half-lives**, so $t_{1/2} = \frac{30.0 \text{ min}}{2} = \mathbf{15.0 \text{ min}}$

(b) $n = \frac{10.0 \text{ d}}{3.82 \text{ d}} = 2.618$; fraction remaining $= \left(\frac{1}{2}\right)^{2.618} = 0.163$, so **16.3 %** remains.

E. Challenge

9. Radiocarbon age.

$$\frac{N_0}{N} = \frac{0.230 \text{ Bq/g}}{0.0850 \text{ Bq/g}} = 2.706 \text{ (the units cancel)}$$
$$n = \frac{\ln 2.706}{\ln 2} = \frac{0.9955}{0.6931} = 1.436 \text{ half-lives}$$
$$t = 1.436 \times 5730 \text{ yr} = \mathbf{8.23 \times 10^3 \text{ yr}}$$
 (about 8230 years). Check: the activity has fallen to 37.0 %, between 50 % and 25 %, so the age lies between 1 and 2 half-lives. ✓